

Optimization for Machine Learning

Course Metadata

Credit Structure	L-T-P: 3-1-0 Total: 4
Offering Semester	4 th Sem, Spring
Level of Subject	UG Level
Compulsory or elective?	Core
Pre-requisite	MA20104
Faculty Names	Jiaul H Paik, Adway Mitra, Swanand Khare
Overlapping Subjects	EE61012 (15%), MA60213 (10%)
Justification of overlap	This course will be a core subject for BTech in AI program

Course Description

Almost all tasks in ML and DL are formulated as an optimization problem where an appropriately chosen loss function is minimized. Because of availability of enormous compute power and large volume of data, very large ML and DL models are being used to solve various learning problems from data. The training of these models is essentially equivalent to solving an optimization problem with a very large number of decision variables. In this course, we aim to develop the foundations of optimization for the students who wish to delve deeper in the theoretical aspects and conduct research in the area of developing new training strategies. The course is designed to achieve the following: i) introduce a student to the theory of optimization, ii) Equip the student with the technical details which will enable them to train ML & DL models.

Course Content

- Motivation and basics of smooth optimization
- Numerical algorithms for optimization problem
- First order and second order methods for optimization
- Scalable optimization techniques based on gradient descent
- Co-ordinate descent and non-smooth optimization

Lecture-wise Breakup

Sl No.	Module		Unit	Module Hr. (48)
1	Motivation	Theory	Formulation of typical ML problems as an optimization problem, Various loss functions in ML/DL, L_2 , L_1 , cross entropy, Hinge and Customized loss functions	3

		Tutorial	Implementation of loss functions and understanding their behavior	1
2	Basics of smooth optimization	Theory	Constrained and unconstrained, convex and non-convex, univariate and multivariate optimization problems, first and second order derivative conditions for local maxima/minima/saddle points, formulae for differentiation with respect to vector, matrices and tensors, Lagrange multiplier theory	6
		Tutorial	Constrained and unconstrained optimization, Auto-differentiation of matrices, vectors and tensors	2
3	First Examples of optimization in ML	Theory	Least squares data fitting (closed form solution), k -means clustering, SVM (with proof)	3
		Tutorial	Optimizing least square and SVM objective functions	1
4	Numerical algorithms for optimization problems	Theory	Need for numerical algorithms: motivating example of training a neural network, first order methods, steepest descent method, step size selection problem	3
		Tutorial	First order numerical method on neural net	1
5	Variation of first order methods	Theory	Adaptive gradient methods (AdaGrad, RMSprop, Adam), momentum-based methods, convergence analysis	6
		Tutorial	Working with adagrad, RMSprop, Adam and momentum with various loss functions	2
6	Addressing scaling up issues	Theory	Parallel and Distributed SGD for large networks, centralized and decentralized SGD, Gradient compression: sparsification and quantization, stability and convergence analysis	4
		Tutorial	Implementation of Parallel, Distributed, centralized and decentralized SGD	1
7	Second order methods	Theory	Newton and Quasi Newton methods, BFGS	3
		Tutorial	Working with Newton method and BFGS	1
8	Coordinate Descent	Theory	Cyclic CD, Randomized CD, Block CD and applications in ML	3

		Tutorial	Optimizing objectives with coordinate descent methods	1
9	Non-smooth optimization	Theory	Calculus of subdifferentials, Optimality conditions for composite non-smooth functions, subgradient descent, Proximal gradient algorithms, Proximal point method	5
		Tutorial	Optimizing composite and non-smooth functions	2

Books

1. **Linear Algebra and Optimization for Machine Learning**, Charu C. Aggarwal, Springer, 2020.
2. **Optimization for Data Analysis**, S. Wright and B. Recht, Cambridge University Press, 2022.
3. **Convex Optimization**, S. Boyd and L. Vandenberghe, Cambridge University Press, 2004.